

Title : AI-Powered Aptitude Assessment & Performance Analytics Platform

Abstract:

The AI-Based Adaptive Aptitude Assessment and Performance Analytics System is a web-based platform developed to enhance placement preparation through intelligent evaluation and analysis of aptitude skills. The system enables students to practice quantitative aptitude, logical reasoning and verbal ability through structured online tests designed according to company-specific patterns. Unlike traditional examination systems, the platform incorporates adaptive difficulty mechanisms that adjust question levels dynamically based on the user's performance, ensuring personalized assessment. The system evaluates not only correctness but also time taken, accuracy rate and topic-wise consistency to generate detailed analytical reports. An integrated analytics engine identifies weak areas, predicts performance trends and recommends targeted practice modules for improvement. Real-time leaderboards and percentile ranking algorithms are implemented to provide competitive benchmarking among users. Additionally, the platform supports secure test environments, randomized question generation and automated result processing to ensure fairness and reliability. Developed using modern web technologies with a secure backend architecture and database management system, the platform ensures scalability, data integrity and efficient performance. The project integrates concepts from data structures, algorithms, statistics, database systems and software engineering principles. By combining adaptive testing with intelligent performance analytics, the system offers a comprehensive solution for placement-oriented aptitude preparation. It aims to assist students in identifying skill gaps, improving problem-solving speed and enhancing overall readiness for competitive recruitment examinations in a structured and data-driven manner.

Keywords: Adaptive Testing, Aptitude Assessment, Performance Analytics, Placement Preparation, Difficulty Adjustment, Percentile Ranking, Weakness Detection, Online Examination System.

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